

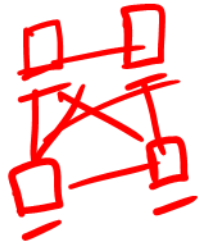
Medium

↓
Guided (wired)

- Twisted Pair Cable
 - STP
 - UTP
- Co-axial cable
- Fiber Optics.

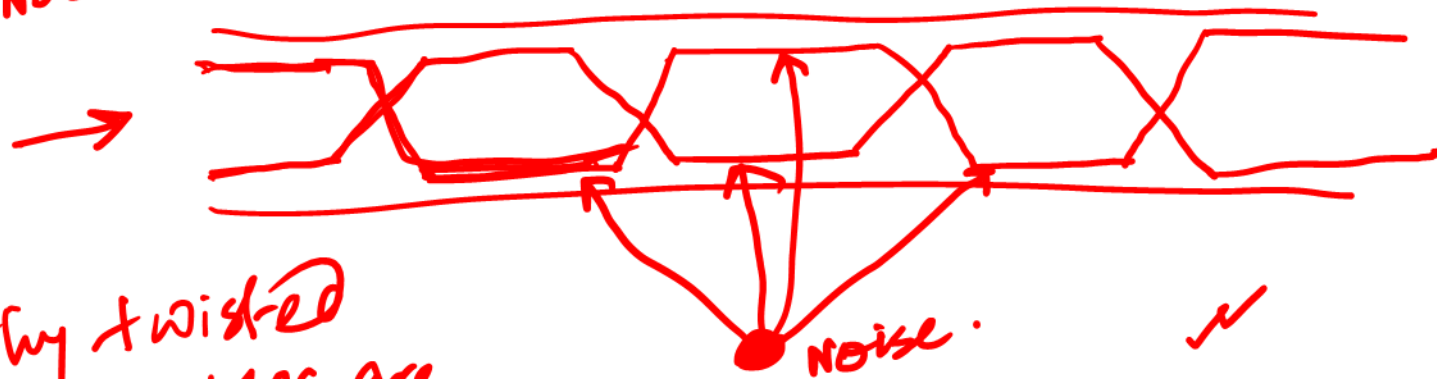
↓
Unguided (wireless).

- Infrared
- Microwave
- Radiowave
- Satellite



Impairments
→ Attenuation
→ Distortion
→ Noise.

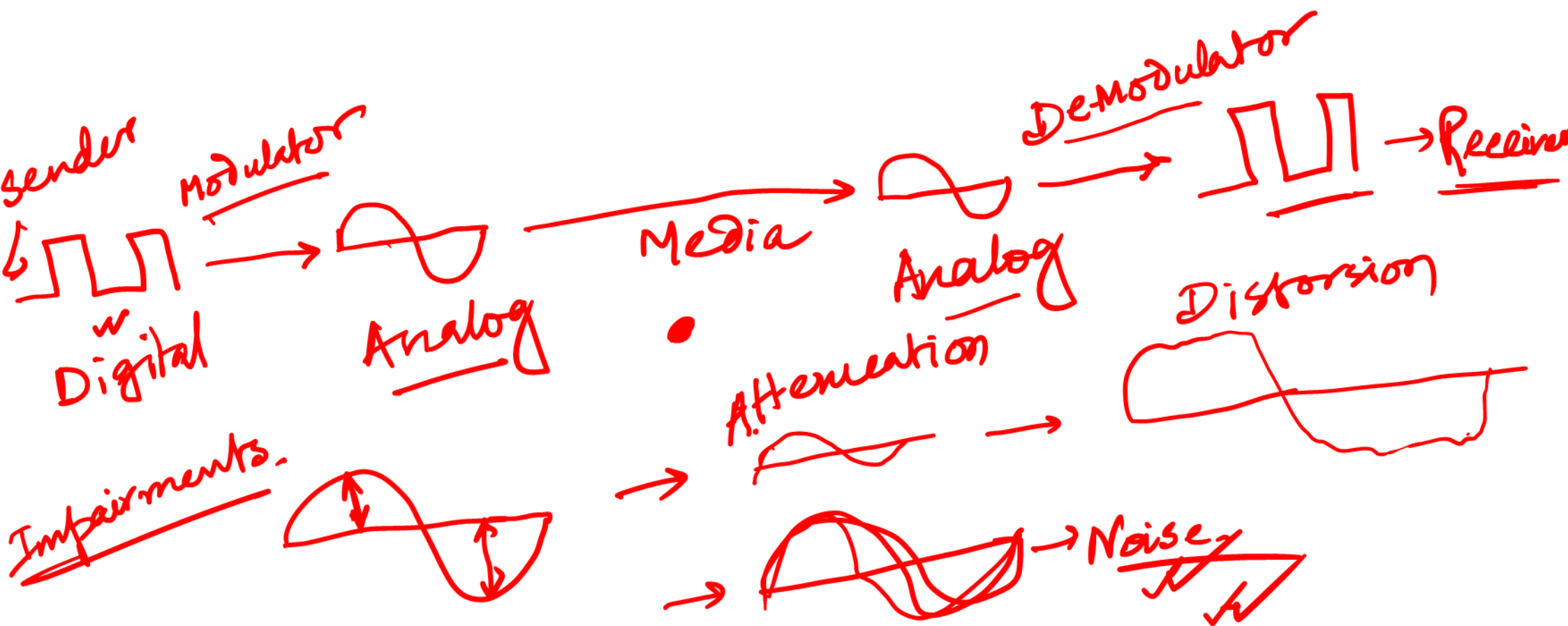
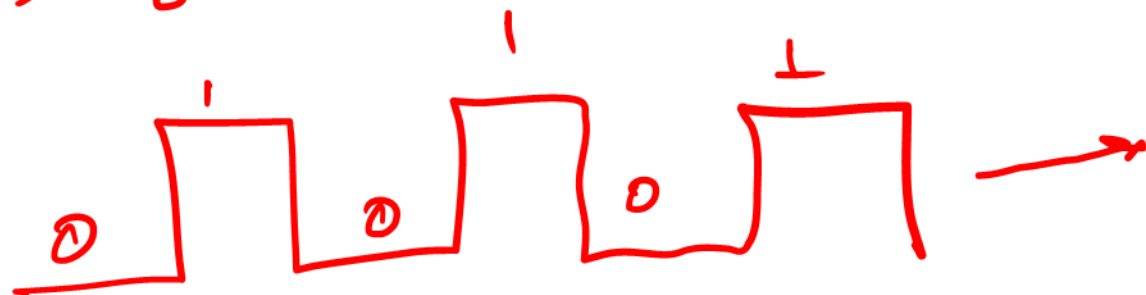
Adv.
Disadv.



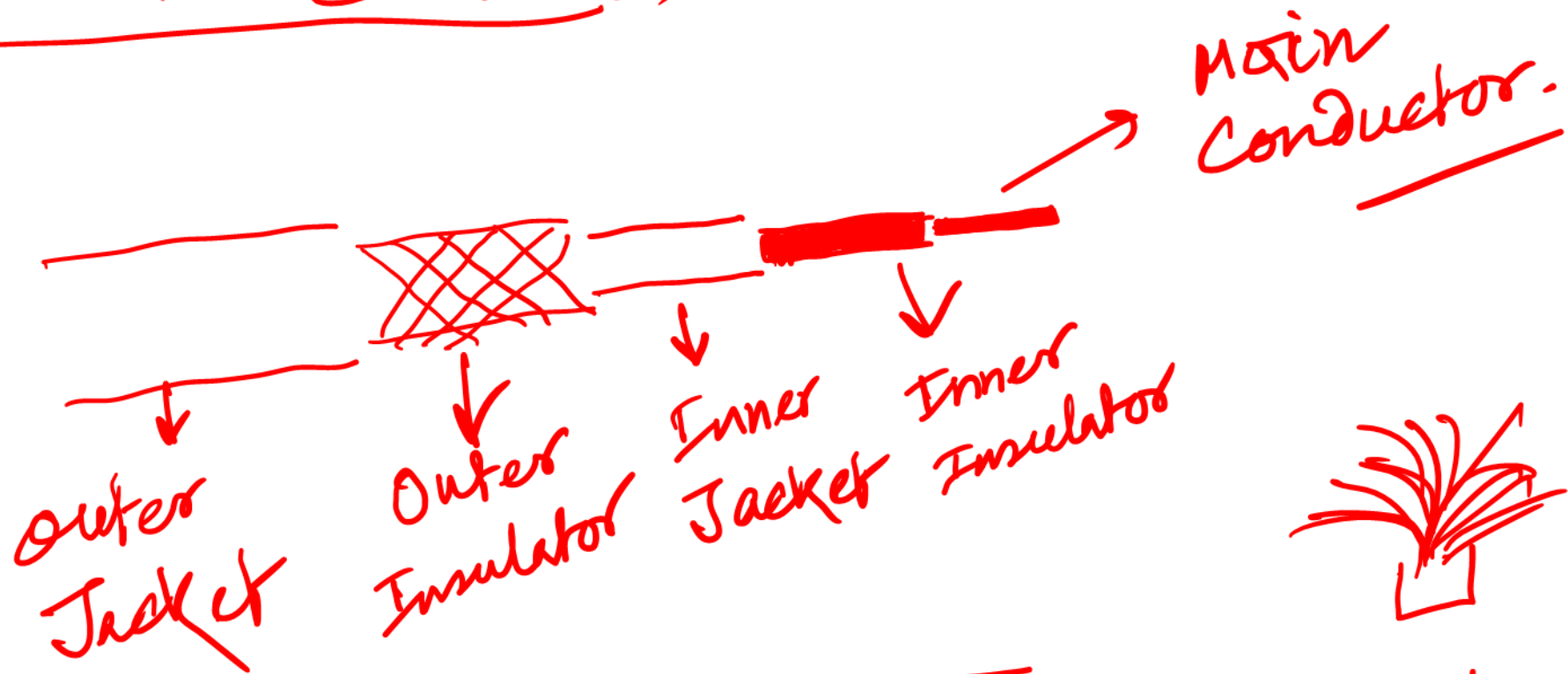
Why twisted
Pair cables are
twisted?



Data ^{data}
→ Analog Transmission
→ Digital data



Co-axial cable :->



Single Layer.

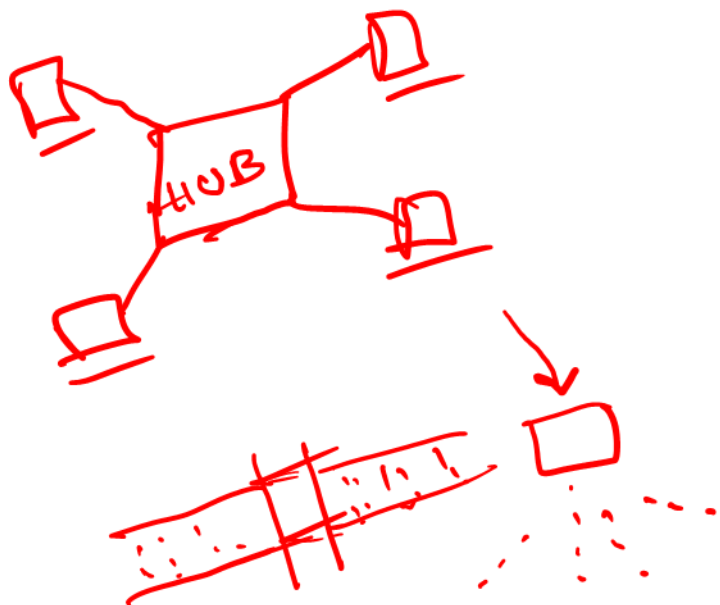
Optical Fibers:
Single Layer Multi-Layer.



Multi-Layer.

Infrared:-





10 Base T
10 Base 2
10 Base 5

Networking Devices.

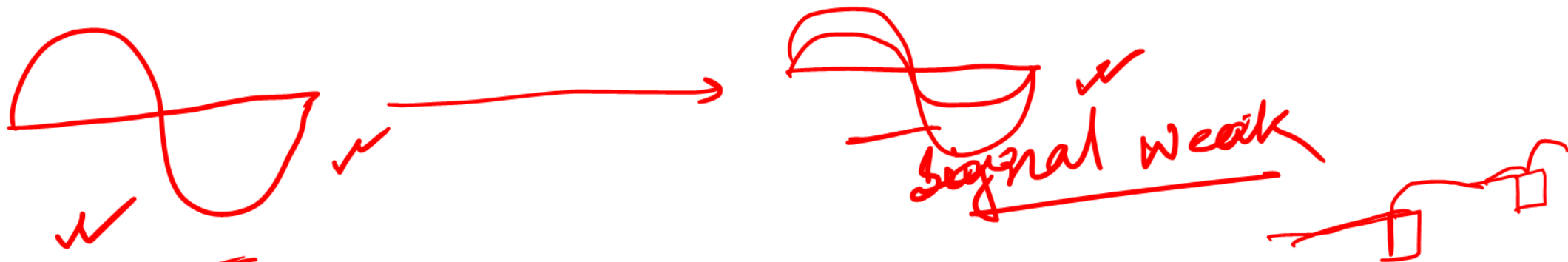
- ✓ ~~MODEM~~ → Modulator - De-Modulator.
D-A A-D.
- HUB
- Switch →
- Bridge →
- Router ✓
- Repeater ✓
- Gateway → Dis-similar Network.
- NIC Card → Physical (46 bit)
- RJ-45 Connector . etc .



10 Base 2
↓
10 Mbps Speed

Base Band Connection

→ 200 meters.
Repeater.



Repeater $\Rightarrow$ Signal Regenerate.
Amplifier $\Rightarrow$ signal Amplify.

